12TH CHEMISTRY · CHAPTER 23 · FULL CHAPTER MCQS

Carbonyl Compounds and Carboxylic Acids

125 topic-wise MCQs from the complete chapter with answers and reasons

23.1 CARBONYL COMPOUNDS

1. The carbonyl group is:

(a) –OH (b) >C=O
(c) –COOH (d) –C≡N

Answer: (b) >C=O — It is present in aldehydes and ketones.

2. The general formula of an aldehyde is:

(a) R–CO–R (b) R–OH
(c) R–COOH (d) R–CHO

Answer: (d) R–CHO — The carbonyl carbon carries a hydrogen atom.

3. The general formula of a ketone is:

(a) R–CO–R (b) R–CHO
(c) R–COOR (d) R–O–R

Answer: (a) R–CO–R — Two alkyl groups on the carbonyl carbon.

4. The most common ketone is:

(a) Butanone (b) Formalin
(c) Acetone (d) Benzaldehyde

Answer: (c) Acetone — Propanone.

5. Acetone is found in:

(a) Nail polish removers (b) Vinegar
(c) Ants (d) Palm oil

Answer: (a) Nail polish removers — Stated in the textbook.

6. The smell and taste of bitter almond is due to:

(a) Acetone (b) Ethyl ethanoate
(c) Butanoic acid (d) Benzaldehyde

Answer: (d) Benzaldehyde — Formed by the oxidation of terpenes.

7. Formalin is an aqueous solution of methanal of strength:

(a) 10 % (b) 40 %
(c) 60 % (d) 100 %

Answer: (b) 40 % — It is used to preserve biological specimens.

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8. At room temperature methanal is a:

(a) Liquid (b) Solid
(c) Gas (d) Crystal

Answer: (c) Gas — Other short-chain aldehydes and ketones are liquids.

9. Aldehydes are prepared by the oxidation of:

(a) Secondary alcohols (b) Primary alcohols
(c) Tertiary alcohols (d) Ketones

Answer: (b) Primary alcohols — With acidified $K_2Cr_2O_7$ or $KMnO_4$.

10. In the preparation of ethanal, the aldehyde is distilled off at once to prevent:

(a) Its reduction
(b) Explosion
(c) Evaporation of ethanol
(d) Its further oxidation to ethanoic acid

Answer: (d) Its further oxidation to ethanoic acid — Aldehydes are easily oxidised.

11. Ethanal can be separated from ethanol by distillation because ethanal has:

(a) A lower boiling point
(b) A higher boiling point
(c) The same boiling point
(d) No boiling point

Answer: (a) A lower boiling point — Ethanol has the higher boiling point.

12. Pure ethanal is obtained by:

(a) Filtration (b) Crystallisation
(c) Redistillation (d) Sublimation

Answer: (c) Redistillation — Stated in the textbook.

13. Oxidation of propan-2-ol gives:

(a) Propanal (b) Propene
(c) Propanoic acid (d) Propanone

Answer: (d) Propanone — A secondary alcohol gives a ketone.

14. Ketones are used in the:

(a) Pharmaceutical, chemical and paint industries
(b) Fertilizer industry
(c) Steel industry
(d) Cement industry

Answer: (a) Pharmaceutical, chemical and paint industries — Stated in the textbook.

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23.2 CHEMISTRY OF CARBONYL COMPOUNDS

15. The carbon atom of the carbonyl group is:

(a) sp^2 hybridized
(b) sp hybridized
(c) sp^3 hybridized
(d) Unhybridized

Answer: (a) sp^2 hybridized — So is the oxygen atom.

16. The carbonyl carbon is:

(a) Nucleophilic
(b) Neutral
(c) Electrophilic
(d) Negatively charged

Answer: (c) Electrophilic — It carries a δ+ charge.

17. The characteristic reactions of aldehydes and ketones are:

(a) Electrophilic additions
(b) Nucleophilic additions
(c) Electrophilic substitutions
(d) Free-radical substitutions

Answer: (b) Nucleophilic additions — Nucleophiles attack the δ+ carbon.

18. The shape around the carbonyl carbon is:

(a) Tetrahedral
(b) Pyramidal
(c) Linear
(d) Planar

Answer: (d) Planar — This makes the carbon uncrowded.

19. The most reactive towards nucleophiles is:

(a) Methanal
(b) Propanone
(c) Ethanal
(d) Butanone

Answer: (a) Methanal — It has no alkyl group.

20. Ketones are less reactive than aldehydes because of:

(a) Hydrogen bonding
(b) Higher mass
(c) Resonance only
(d) Inductive effect and steric hindrance

Answer: (d) Inductive effect and steric hindrance — Two alkyl groups reduce δ+ and block the nucleophile.

21. Alkyl groups attached to the carbonyl carbon are:

(a) Electron-withdrawing
(b) Neutral
(c) Electron-releasing
(d) Acidic

Answer: (c) Electron-releasing — They reduce the positive charge on carbon.

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22. The getting in the way of the nucleophile by alkyl groups is called:

(a) Inductive effect
(b) Steric hindrance
(c) Resonance
(d) Polarity

Answer: (b) Steric hindrance — It is greatest in ketones.

23. In the carbonyl group, oxygen carries:

(a) δ+ charge
(b) A full positive charge
(c) No charge
(d) δ– charge

Answer: (d) δ– charge — It is more electronegative than carbon.

24. Which is the least reactive towards nucleophilic attack?

(a) $(CH_3)_2CO$
(b) $(CH_3)_3CCOC(CH_3)_3$
(c) CH_3CHO
(d) HCHO

Answer: (b) $(CH_3)_3CCOC(CH_3)_3$ — It has the bulkiest groups.

23.3 NUCLEOPHILIC ADDITION REACTIONS

25. Aldehydes and ketones react with HCN to form:

(a) Alcohols
(b) Hydrazones
(c) Esters
(d) Cyanohydrins

Answer: (d) Cyanohydrins — Also called hydroxynitriles.

26. HCN is not used directly because it is:

(a) An extremely poisonous gas
(b) Expensive
(c) Unreactive
(d) A solid

Answer: (a) An extremely poisonous gas — It is generated from KCN and HCl.

27. Ethanal with HCN gives:

(a) Propanenitrile
(b) 2-Hydroxypropanenitrile
(c) Ethanol
(d) Ethanoic acid

Answer: (b) 2-Hydroxypropanenitrile — A hydroxynitrile.

28. Propanone with HCN gives:

(a) 2-Hydroxypropanenitrile
(b) Propan-2-ol
(c) 2-Hydroxy-2-methylpropanenitrile
(d) Propanoic acid

Answer: (c) 2-Hydroxy-2-methylpropanenitrile — Used in the manufacture of Perspex.

29. The nucleophile in the addition of HCN is:

(a) H^+ (b) K^+
(c) HCN (d) CN^-

Answer: (d) CN^- — It attacks the carbonyl carbon.

30. In the reaction of HCN with carbonyl compounds, KCN is used in:

(a) Limiting amount (b) Slight excess
(c) Equal amount (d) No amount

Answer: (b) Slight excess — To provide free CN^- ions.

31. Hydrolysis of nitriles gives:

(a) Carboxylic acids (b) Alcohols
(c) Ketones (d) Amines

Answer: (a) Carboxylic acids — Or carboxylate ions in alkali.

32. $NaBH_4$ reduces an aldehyde to a:

(a) Secondary alcohol (b) Carboxylic acid
(c) Primary alcohol (d) Ketone

Answer: (c) Primary alcohol — E.g. ethanal to ethanol.

33. $NaBH_4$ reduces a ketone to a:

(a) Secondary alcohol (b) Primary alcohol
(c) Tertiary alcohol (d) Aldehyde

Answer: (a) Secondary alcohol — E.g. propanone to propan-2-ol.

34. The nucleophile supplied by the reducing agent in reduction is the:

(a) Hydrogen ion (b) Hydride ion
(c) Hydroxide ion (d) Hydrogen atom

Answer: (b) Hydride ion — H^-.

35. The intermediate in the reduction of a carbonyl compound is:

(a) A carbocation (b) A carbanion
(c) A free radical (d) An alkoxide ion

Answer: (d) An alkoxide ion — It takes a proton from water.

36. [H] in a reduction equation represents:

(a) Atomic hydrogen
(b) Hydrogen gas
(c) Reducing hydrogen from the reducing agent
(d) A proton

Answer: (c) Reducing hydrogen from the reducing agent — It only balances the equation.

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37. 2,4-DNPH with a carbonyl compound gives:

(a) A silver mirror
(b) An orange crystalline solid
(c) A brick-red precipitate
(d) A yellow precipitate

Answer: (b) An orange crystalline solid — A 2,4-dinitrophenylhydrazone.

38. The reaction of 2,4-DNPH with a carbonyl compound is:

(a) Substitution
(b) Reduction
(c) Oxidation
(d) Condensation (addition–elimination)

Answer: (d) Condensation (addition–elimination) — Water is eliminated.

39. The small molecule lost in the 2,4-DNPH reaction is:

(a) HCl (b) Ammonia
(c) Water (d) Hydrogen

Answer: (c) Water — It is a condensation reaction.

40. Which compound gives the 2,4-DNPH test?

(a) Propanone (b) Ethanoic acid
(c) Ethyl ethanoate (d) Ethanol

Answer: (a) Propanone — Only aldehydes and ketones give it.

41. The 2,4-dinitrophenylhydrazones are useful because they have:

(a) Sweet smells
(b) Low boiling points
(c) Well-defined melting points
(d) A blue colour

Answer: (c) Well-defined melting points — This helps to identify the carbonyl compound.

23.3.2 TEST OF ALDEHYDES AND KETONES

42. Ketones resist oxidation because:

(a) They are saturated (b) They are gases
(c) They are non-polar (d) A C–C bond has to be broken

Answer: (d) A C–C bond has to be broken — Only strong oxidising agents oxidise them.

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43. Tollens' reagent is:

(a) Ammoniacal silver nitrate (b) Alkaline copper(II) tartrate
(c) Acidified dichromate (d) Iodine in alkali

Answer: (a) Ammoniacal silver nitrate — It contains silver ions in alkaline solution.

44. A positive Tollens' test is shown by:

(a) A yellow precipitate (b) A silver mirror
(c) A brick-red precipitate (d) Orange crystals

Answer: (b) A silver mirror — Silver ions are reduced to silver.

45. Tollens' test is given by:

(a) Ketones (b) Alcohols
(c) Aldehydes (d) Esters

Answer: (c) Aldehydes — Ketones do not give it.

46. Mirrors for domestic use were made using Tollens' reagent and:

(a) Ethanol (b) Ethanoic acid
(c) Acetone (d) Glucose

Answer: (d) Glucose — An aldehydic sugar.

47. The complex ion in Tollens' reagent is:

(a) $[Cu(NH_3)_4]^{2+}$
(b) $[Ag(CN)_2]^-$
(c) $[Ag(NH_3)_2]^+$
(d) $[Ag(H_2O)_2]^+$

Answer: (c) $[Ag(NH_3)_2]^+$ — In $[Ag(NH_3)_2]OH$.

48. Fehling's solution is:

(a) Bright blue (b) Colourless
(c) Orange (d) Purple

Answer: (a) Bright blue — Due to Cu^{2+} ions.

49. In Fehling's solution, copper(II) ions are complexed with a salt of:

(a) Citric acid (b) Tartaric acid
(c) Ethanoic acid (d) Oxalic acid

Answer: (b) Tartaric acid — In an alkaline medium.

50. The brick-red precipitate in Fehling's test is:

(a) CuO (b) Cu
(c) $Cu(OH)_2$ (d) Cu_2O

Answer: (d) Cu_2O — Copper(I) oxide.

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51. In Fehling's test the aldehyde is:

(a) Oxidised (b) Reduced
(c) Hydrolysed (d) Dehydrated

Answer: (a) Oxidised — Cu^{2+} is reduced to Cu^{+}.

52. The iodoform test identifies the group:

(a) –CHO (b) $CH_3CO–$
(c) –COOH (d) –OH

Answer: (b) $CH_3CO–$ — The acetyl (ethanoyl) group.

53. Iodoform is:

(a) CH_3I (b) CI_4
(c) CHI_3 (d) CH_2I_2

Answer: (c) CHI_3 — Triiodomethane.

54. Iodoform is a:

(a) White liquid (b) Silver mirror
(c) Red precipitate (d) Yellow crystalline solid

Answer: (d) Yellow crystalline solid — Stated in the textbook.

55. The only aldehyde that gives the iodoform test is:

(a) Ethanal (b) Methanal
(c) Propanal (d) Benzaldehyde

Answer: (a) Ethanal — It has the $CH_3CO–$ group.

56. The reagents for the iodoform test are:

(a) Br_2 and H_2O
(b) $AgNO_3$ and NH_3
(c) I_2 and NaOH
(d) $K_2Cr_2O_7$ and H_2SO_4

Answer: (c) I_2 and NaOH — Alkaline iodine.

57. Which alcohol gives the iodoform test?

(a) Methanol (b) Ethanol
(c) Propan-1-ol (d) 2-Methylpropan-2-ol

Answer: (b) Ethanol — It has the $CH_3–CH(OH)–$ type structure.

58. Which ketone does NOT give the iodoform test?

(a) Propanone (b) Pentan-2-one
(c) Butanone (d) Pentan-3-one

Answer: (d) Pentan-3-one — It is not a methyl ketone.

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59. Propanal and propanone can be distinguished by:

(a) 2,4-DNPH (b) Tollens' reagent
(c) Sodium metal (d) Litmus

Answer: (b) Tollens' reagent — Only propanal gives a silver mirror.

23.4 CARBOXYLIC ACIDS AND THEIR PREPARATION

60. The functional group of carboxylic acids is:

(a) –COOH (b) –CHO
(c) –COCl (d) –OH

Answer: (a) –COOH — The carboxyl group.

61. Methanoic acid is found in:

(a) Vinegar (b) Palm oil
(c) Butter (d) Ants

Answer: (d) Ants — It is formic acid.

62. The acid present in vinegar is:

(a) Methanoic acid (b) Butanoic acid
(c) Ethanoic acid (d) Palmitic acid

Answer: (c) Ethanoic acid — Acetic acid.

63. Rancid butter contains:

(a) Formic acid (b) Butanoic acid
(c) Oleic acid (d) Benzoic acid

Answer: (b) Butanoic acid — Butyric acid.

64. The melting point of ethanoic acid is:

(a) 0 °C (b) 118 °C
(c) 77 °C (d) 17 °C

Answer: (d) 17 °C — So it solidifies in winter.

65. The boiling point of ethanoic acid is:

(a) 118 °C (b) 17 °C
(c) 77 °C (d) 100 °C

Answer: (a) 118 °C — High because of hydrogen bonding.

66. The solid form of pure ethanoic acid is called:

(a) Formalin (b) Vinegar
(c) Glacial acetic acid (d) Dry ice

Answer: (c) Glacial acetic acid — It resembles glaciers.

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67. Ethanoic acid exists as a dimer because of:

(a) Ionic bonds
(b) Hydrogen bonding
(c) Covalent bonds
(d) Metallic bonds

Answer: (b) Hydrogen bonding — Two hydrogen bonds hold the pair.

68. Long-chain aliphatic monocarboxylic acids are called:

(a) Fatty acids
(b) Amino acids
(c) Mineral acids
(d) Esters

Answer: (a) Fatty acids — E.g. stearic, palmitic and oleic acids.

69. Fatty acids occur in vegetable oils as:

(a) Free acids
(b) Salts
(c) Triglycerides
(d) Amides

Answer: (c) Triglycerides — Stated in the textbook.

70. Palmitic acid is found in large quantities in:

(a) Vinegar
(b) Palm oil
(c) Ants
(d) Almonds

Answer: (b) Palm oil — Its formula is $C_{16}H_{32}O_2$.

71. To prepare a carboxylic acid, a primary alcohol is:

(a) Distilled with limited oxidant
(b) Reduced
(c) Heated with NaOH
(d) Refluxed with excess oxidising agent

Answer: (d) Refluxed with excess oxidising agent — So the aldehyde is oxidised further.

72. Acid hydrolysis of ethanenitrile gives ethanoic acid and:

(a) NH_3
(b) NH_4Cl
(c) NaCl
(d) N_2

Answer: (b) NH_4Cl — $CH_3CN + 2H_2O + HCl$.

73. Alkaline hydrolysis of a nitrile first gives:

(a) The free acid
(b) An ester
(c) An alcohol
(d) The salt of the acid

Answer: (d) The salt of the acid — The acid is obtained on acidification.

74. Methylbenzene is oxidised by acidified potassium manganate(VII) to:

(a) Benzaldehyde
(b) Phenol
(c) Benzoic acid
(d) Benzene

Answer: (c) Benzoic acid — Benzenecarboxylic acid.

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75. Alkaline hydrolysis of ethyl ethanoate gives ethanol and:

(a) Sodium ethanoate (b) Ethanoic acid
(c) Ethanal (d) Sodium ethoxide

Answer: (a) Sodium ethanoate — The carboxylate salt.

23.5 REACTIONS OF CARBOXYLIC ACIDS

76. Ethanoic acid reacts with sodium to give sodium ethanoate and:

(a) Water (b) Oxygen
(c) Carbon dioxide (d) Hydrogen

Answer: (d) Hydrogen — A redox reaction with an active metal.

77. Ethanoic acid with NaOH gives:

(a) An ester (b) A salt and water
(c) An acyl chloride (d) An alcohol

Answer: (b) A salt and water — Neutralisation.

78. Carboxylic acids liberate carbon dioxide from:

(a) Carbonates and bicarbonates (b) Alkalis
(c) Active metals (d) Alcohols

Answer: (a) Carbonates and bicarbonates — A salt and water are also formed.

79. Ethanoic acid with thionyl chloride gives:

(a) Chloroethane (b) Chloroethanoic acid
(c) Ethanoyl chloride (d) Ethyl ethanoate

Answer: (c) Ethanoyl chloride — With SO_2 and HCl.

80. The by-products of the reaction of ethanoic acid with PCl_5 are:

(a) SO_2 and HCl
(b) HCl and $POCl_3$
(c) H_3PO_3 only
(d) H_2O and Cl_2

Answer: (b) HCl and $POCl_3$ — Given in the textbook.

81. The catalyst in esterification is:

(a) Concentrated H_2SO_4
(b) Dilute NaOH
(c) Al_2O_3
(d) $LiAlH_4$

Answer: (a) Concentrated H_2SO_4 — The reaction is reversible.

82. Ethanoic acid heated with ethanol and concentrated sulphuric acid gives:

(a) Ethyl methanoate (b) Methyl ethanoate
(c) Ethyl ethanoate (d) Ethanoyl chloride

Answer: (c) Ethyl ethanoate — Esterification.

83. Carboxylic acids are reduced to primary alcohols by:

(a) $NaBH_4$
(b) $K_2Cr_2O_7$
(c) H_2/Ni
(d) $LiAlH_4$

Answer: (d) $LiAlH_4$ — The other reducing agents do not work.

84. Reduction of ethanoic acid gives:

(a) Ethanal only (b) Ethane
(c) Ethanol (d) Methanol

Answer: (c) Ethanol — A primary alcohol.

85. The only monocarboxylic acid giving a silver mirror test is:

(a) Ethanoic acid (b) Methanoic acid
(c) Propanoic acid (d) Benzoic acid

Answer: (b) Methanoic acid — It has an aldehydic portion.

86. The reducing behaviour of methanoic acid is due to its:

(a) Carboxylic portion (b) Hydrogen bonding
(c) Hydroxyl group (d) Aldehydic portion

Answer: (d) Aldehydic portion — –CHO part of H–CO–OH.

87. Methanoic acid with Fehling's solution gives:

(a) A brick-red precipitate (b) A silver mirror
(c) A yellow precipitate (d) No change

Answer: (a) A brick-red precipitate — Cu_2O.

88. Methanoic acid is oxidised to:

(a) Methanal (b) Carbon dioxide and water
(c) Carbon monoxide (d) Methanol

Answer: (b) Carbon dioxide and water — By weak and strong oxidising agents.

89. Ethanedioic acid with warm acidified potassium manganate(VII) gives:

(a) Carbon dioxide and water (b) Ethanoic acid
(c) Ethanol (d) Methanoic acid

Answer: (a) Carbon dioxide and water — It is completely oxidised.

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90. Oxidation is the:

(a) Gain of electrons
(b) Loss of oxygen
(c) Gain of hydrogen
(d) Loss of electrons

Answer: (d) Loss of electrons — With an increase in oxidation state.

91. The formula of ethanedioic acid is:

(a) CH_3COOH
(b) HCOOH
(c) HOOC–COOH
(d) C_6H_5COOH

Answer: (c) HOOC–COOH — It has two carboxyl groups.

23.6 RELATIVE ACIDITIES

92. The correct order of acid strength is:

(a) Alcohols > phenols > carboxylic acids
(b) Alcohols > carboxylic acids > phenols
(c) Phenols > carboxylic acids > alcohols
(d) Carboxylic acids > phenols > alcohols

Answer: (d) Carboxylic acids > phenols > alcohols — Given in the textbook.

93. The approximate K_a of a carboxylic acid is:

(a) 10^{-5}
(b) 10^{-16}
(c) 10^{-10}
(d) 10^{-14}

Answer: (a) 10^{-5} — Table 23.1.

94. The approximate pK_a of a phenol is:

(a) 5
(b) 14
(c) 10
(d) 16

Answer: (c) 10 — Table 23.1.

95. The approximate pK_a of an alcohol is:

(a) 5
(b) 16
(c) 10
(d) 14

Answer: (b) 16 — Table 23.1.

96. pK_a is equal to:

(a) $-\log K_a$
(b) $\log K_a$
(c) $1/K_a$
(d) K_a^2

Answer: (a) $-\log K_a$ — Definition.

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97. A stronger acid has:

(a) A higher pK_a (b) No K_a
(c) A lower K_a (d) A lower pK_a

Answer: (d) A lower pK_a — Higher K_a, lower pK_a.

98. The ethanoate ion is stable because of:

(a) Hydrogen bonding
(b) +I effect
(c) Delocalisation of the negative charge
(d) Its large size

Answer: (c) Delocalisation of the negative charge — Over the –COO group.

99. An electron-donating group makes a carboxylic acid:

(a) Stronger (b) Weaker
(c) Neutral (d) Unchanged

Answer: (b) Weaker — It intensifies the charge on the carboxylate ion.

100. Methanoic acid, compared with ethanoic acid, is about:

(a) 10 times weaker (b) 100 times stronger
(c) 10 times stronger (d) Equally strong

Answer: (c) 10 times stronger — The methyl group has a +I effect.

101. The pK_a of chloroacetic acid is:

(a) 4.74 (b) 2.86
(c) 1.26 (d) 0.64

Answer: (b) 2.86 — Given in the textbook.

102. The pK_a of trichloroacetic acid is:

(a) 2.86 (b) 4.74
(c) 1.26 (d) 0.64

Answer: (d) 0.64 — It is comparable to some mineral acids.

103. The strongest acid among these is:

(a) Cl_3CCOOH (b) CH_3COOH
(c) $ClCH_2COOH$ (d) $Cl_2CHCOOH$

Answer: (a) Cl_3CCOOH — Three electron-withdrawing chlorine atoms.

104. Chlorine increases the acidity of a carboxylic acid by its:

(a) +I effect (b) Hydrogen bonding
(c) Steric effect (d) –I effect

Answer: (d) –I effect — It disperses the negative charge.

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23.7 – 23.9 ESTERS AND ACYL CHLORIDES

105. Esters have lower boiling points than the corresponding alcohols because they:

(a) Are ionic
(b) Are aromatic
(c) Are heavier
(d) Cannot form hydrogen bonds with each other

Answer: (d) Cannot form hydrogen bonds with each other — Stated in the textbook.

106. The boiling point of ethyl ethanoate is:

(a) 17 °C (b) 77 °C
(c) 118 °C (d) 100 °C

Answer: (b) 77 °C — Given in the textbook.

107. The fragrance and taste of fruits are due to:

(a) Aldehydes (b) Carboxylic acids
(c) Esters (d) Ketones

Answer: (c) Esters — They have sweet fruity smells.

108. Esters are used as:

(a) Solvents and flavouring agents (b) Antiseptics
(c) Fuels (d) Oxidising agents

Answer: (a) Solvents and flavouring agents — In fruit drinks and sweets.

109. The reaction of a carboxylic acid with an alcohol in acidic medium is called:

(a) Saponification
(b) Condensation polymerisation
(c) Hydrolysis
(d) Fischer’s esterification

Answer: (d) Fischer’s esterification — Named in the textbook.

110. Propanoic acid with ethanol gives:

(a) Ethyl ethanoate (b) Propyl ethanoate
(c) Ethyl propanoate (d) Methyl propanoate

Answer: (c) Ethyl propanoate — Acid part: propanoate; alcohol part: ethyl.

111. The reaction of an acyl chloride with an alcohol:

(a) Is reversible (b) Goes to completion
(c) Needs conc. sulphuric acid (d) Does not occur

Answer: (b) Goes to completion — No equilibrium mixture is formed.

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112. Ethanoyl chloride with ethanol gives ethyl ethanoate and:

(a) HCl
(b) Water
(c) Cl_2
(d) SO_2

Answer: (a) HCl — The reaction is vigorous.

113. Phenyl esters are prepared from phenol and:

(a) Carboxylic acids
(b) Aldehydes
(c) Acyl chlorides
(d) Alcohols

Answer: (c) Acyl chlorides — On warming.

114. Hydrolysis of ethyl benzoate gives benzoic acid and:

(a) Methanol
(b) Benzene
(c) Phenol
(d) Ethanol

Answer: (d) Ethanol — Acid- or base-catalysed.

115. Hydrolysis of methyl benzoate gives:

(a) Benzoic acid and ethanol
(b) Benzoic acid and methanol
(c) Phenol and ethanoic acid
(d) Benzene and methanol

Answer: (b) Benzoic acid and methanol — Aromatic esters hydrolyse in the same way.

116. The functional group of acyl chlorides is:

(a) –COCl
(b) –Cl
(c) –COOH
(d) –CHO

Answer: (a) –COCl — They are named alkanoyl halides.

117. The general formula of saturated aliphatic acyl chlorides is:

(a) $C_nH_{2n+1}Cl$
(b) $C_nH_{2n+1}COOH$
(c) $C_nH_{2n}COCl$
(d) $C_nH_{2n+1}COCl$

Answer: (d) $C_nH_{2n+1}COCl$ — Given in the textbook.

118. CH_3–CH_2–COCl is named:

(a) Propanoyl chloride
(b) Ethanoyl chloride
(c) Chloropropane
(d) Butanoyl chloride

Answer: (a) Propanoyl chloride — Three carbon atoms.

119. The most reactive derivatives of carboxylic acids are:

(a) Esters
(b) Acyl chlorides
(c) Amides
(d) Salts

Answer: (b) Acyl chlorides — So other derivatives are made from them.

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120. Ethanoyl chloride is hydrolysed by water:

(a) Only on reflux (b) Never
(c) Rapidly at room temperature (d) In 14 days

Answer: (c) Rapidly at room temperature — Its carbon is strongly δ+.

121. The hydrolysis of chloroethane by water at 100 °C takes about:

(a) 14 days (b) A few minutes
(c) One hour (d) One year

Answer: (a) 14 days — Stated in the textbook.

122. Chlorobenzene with water gives:

(a) Phenol rapidly (b) No reaction
(c) Benzene (d) Benzoic acid

Answer: (b) No reaction — Its C–Cl bond has double-bond character.

123. The correct order of ease of hydrolysis is:

(a) Aryl > alkyl > acyl chloride
(b) Acyl > aryl > alkyl chloride
(c) Alkyl > acyl > aryl chloride
(d) Acyl > alkyl > aryl chloride

Answer: (d) Acyl > alkyl > aryl chloride — Section 23.9.

124. The carbonyl carbon of ethanoyl chloride is strongly δ+ because it is attached to:

(a) Two carbon atoms (b) Hydrogen only
(c) Oxygen and chlorine (d) A benzene ring

Answer: (c) Oxygen and chlorine — Both are highly electronegative.

125. Hydrolysis of ethanoyl chloride gives ethanoic acid and:

(a) Cl_2 (b) HCl
(c) H_2 (d) NaCl

Answer: (b) HCl — At room temperature.

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